

Supplementary materials

of “Cardiac 3D Motion Reconstruction Using Dual-Camera Defocused Speckle Imaging with Multi-Scale Amplification”

<https://github.com/contactless-healthcare/Dual-Camera-for-3DSCG-GCG-Reconstruction>

1 Summary of related research on cardiac activity monitoring

Recent research on cardiac activity monitoring encompasses both contact-based and non-contact-based approaches, each offering distinct advantages and limitations. In this section, we supplemented the details of some recent research, including measurement devices, datasets, sensor placement, and applications. Table 1 summarizes studies on contact-based cardiac activity monitoring, while Table 2 presents studies on non-contact-based approaches. Furthermore, we compare contact-based sensors, radar-based, camera-based, and defocused camera-based cardiac monitoring technologies in terms of contact-free, spatial resolution, motion sensitivity, system cost, signal fidelity, signal dimension and the need for alignment. Detailed information is presented in Table 3.

Table 1: Detailed information on recent studies of contact-based cardiac activity monitoring.

Modality	Paper	Measurement Device	Dataset	Location	Application
SCG	Ganti <i>et al.</i> [1]	Tri-axial Accelerometer.	45 congenital heart disease.	sternum.	Estimation of stroke volume.
	Shandhi <i>et al.</i> [2]	Wearable patch (500 Hz).	20 heart failure.	sternum.	Estimation of PAM and PCWP.
	Singh <i>et al.</i> [3]	LLA SCG sensor.	100 patients.	chest.	Assessment of hemodynamic parameters before and after transcatheter aortic valve implantation.
GCG	Presti <i>et al.</i> [4]	Gyroscopes (60 Hz).	1 healthy.	chest.	Estimation of HR and RR.
	Sieciński <i>et al.</i> [5]	Gyroscopes (800 Hz).	29 healthy.	sternum.	Estimation of HRV.
	Jafari Tadi <i>et al.</i> [6]	Inertial measurement unit (800 Hz).	9 healthy.	chest.	Estimation of hemodynamic parameters and timings of peak myocardial movements and deformations.
SCG & GCG	Mehrang <i>et al.</i> [7]	Smartphone (200 Hz).	300 patients with AF or HF.	chest.	Multilabel and Stratified Classification of AF and HF.
	Yang <i>et al.</i> [8]	Accelerometer & Gyroscope (256 Hz).	20 healthy & 20 aortic stenosis.	sternum.	Detection of AS and classification valvular heart diseases using time-frequency features of SCG and GCG.
	Tapotee <i>et al.</i> [9]	Accelerometer & Gyroscope (800 Hz).	29 healthy & 12 patients.	chest.	Reconstruction of ECG using SCG and GCG.

Table 2: **Detailed information on recent studies of Non-contact based cardiac activity monitoring.**

Modality	Paper	Dataset	Application	Method
RF -based	Xia <i>et al.</i> [10]	8 healthy.	Waveform Comparison	SCG monitoring using Doppler radar and comparison with contact-SCG.
	Ha <i>et al.</i> [11]	40 healthy.	Biomarkers recognition.	Reconstructing SCG waveforms and detecting the timing of 5 key cardiovascular events (AO,AC,MO,MC,IC) using deep learning techniques
	Li <i>et al.</i> [12]	3 healthy.	HR, RR monitoring.	Employed variational mode separation to separate respiratory and cardiac motion patterns from 4D radar signals.
	Chen <i>et al.</i> [13]	35 healthy.	ECG signal reconstruct	Development of a millimeter-wave radar system to improve cardiac activity detection accuracy and reconstruct ECG signal.
	Wang <i>et al.</i> [14]	11 healthy.	BBI, HRV monitoring.	HRV monitoring based on commercial millimeter wave radio signals.
Camera -based	Rahman et al. [15]	14 healthy.	HR monitoring & Waveform Comparison.	Estimation of SCG signals by tracking the motion of an artificial target (i.e. QR codes attached to the chest
	Rahman et al. [16]	14 healthy.	Multichannel SCG & HR monitoring.	Extracted multi-channel SCG from multiple chest-mounted markers and utilized to analyze chest vibration patterns and estimate HR.
Defocused Camera based	Que et al. [17]	9 healthy.	HR monitoring.	Measurement of single-point SCG signals on the chest surface using a defocused speckle imaging.
	Zhou et al. [18]	21 healthy.	LVET monitoring.	extracted LVET from camera SCG based on a defocused speckle imaging system.
	Liu et al. [19]	19 healthy.	Multi-Point SCG & BP monitoring.	Introduction of a novel concept for contactless monitoring of multi-point cardiac motion using dual-wavelength defocused speckle imaging system.
	Zhu et al. [20]	19 healthy.	HR, HRV, RR, BP monitoring.	Explored the potential of a camera-based dual-modality PPG-SCG system for privacy-preserving vital sign monitoring during sleep.

Table 3: **Comparison of cardiac motion monitoring techniques.**

Technology	Accelerometer /Gyroscope	RF-Based	Camera	Defocused Camera
Contact-Free	No	Yes	Yes	Yes
Spatial Resolution	Moderate	Low	Moderate	High
Motion Sensitivity	High	Low	Moderate	High
System Cost	Low	High	Moderate	Moderate
Signal Fidelity	High	Moderate	Moderate	High
Signal Dimension	3D	1D/2D	2D	2D/3D
Alignment	Need	No need	No need	Need

2 Algorithm of signal segmentation

The period segmentation pseudocode is shown in the following.

Algorithm 1 Algorithm of signal segmentation.

Input: Reconstructed and raw speckle motion signal: $data = [x_1, x_2, \dots, x_n]$ and the ECG signal: ECG .

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1:  $Pk_{data} \leftarrow findpeaks(data(1 : 60 \times 250))$ ,  $Pk_{ecg} \leftarrow findpeaks(ECG)$ .  $\Leftarrow$  Peaks position
   coordinates.
2:  $offset = 0$ ,  $corrmax = -1$ ,  $IBI_{\sim} \leftarrow diff(Pk_{\sim})$ ,  $Pksn_{\sim} \leftarrow length(Pk_{\sim})$ .  $\sim$  denotes ecg or
   data.
3: for each  $i \in [1, Pksn_{data} - Pksn_{ecg}]$  do
4:    $r \leftarrow corr(IBI_{ecg}(i : i + Pksn_{data}), IBI_{data})$ .
5:   if  $r > corrmax$  then
6:      $offset \leftarrow i$ ,  $corrmax \leftarrow r$ .
7:   end if
8: end for
9:  $ECG_{new} \leftarrow ECG(Pk_{ecg}(offset) - Pk_{data}(1) : end)$ ,  $Pk'_{ecg} \leftarrow findpeaks(ECG_{new})$ .
10:  $AO \leftarrow max(data(Pk'_{ecg} : Pk'_{ecg} + 30))$ .
11:  $IBI'_{data} \leftarrow diff(AO)$ .
12: for each  $pks \in [1, length(AO)]$  do
13:    $signal_{pks} \leftarrow data(AO(pks + 1) - 0.3 \times IBI'_{data}(pks) : AO(pks) + 0.7 \times IBI'_{data}(pks + 1))$ 
14: end for

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Output: $signal \Leftarrow$ The single cycle signal after segmentation.

3 Experimental Protocols and System Parameters

3.1 Experimental Protocols

In the laboratory setting, participants lay supine on a bed while synchronized data were collected using two cameras, an accelerometer and a patient monitor. Each recording session lasted 6 minutes. At the 3-minute mark, an ice pack was applied to the participant's palm for 40 seconds to induce cardiovascular changes, then removed to allow recovery. Participants were instructed to minimize movement and repeat the procedure 5 times. In the ICU setting, participants also lay supine, but no cold stimulus was applied. The cameras, inertial sensor, and patient monitor synchronously recorded approximately 10 minutes of data per subject. A laser rangefinder was used during each experiment to capture the precise spatial configuration of the setup for subsequent motion decoupling calculations.

3.2 System Parameters

In the laboratory setting, cameras were positioned 1.2 m above the bed, with focus planes set at 10 cm and 5 cm from the lens. In the ICU, the same focus plane configuration was used, and the distance from the camera lens to the subject's chest was measured using a laser rangefinder (DL4168).

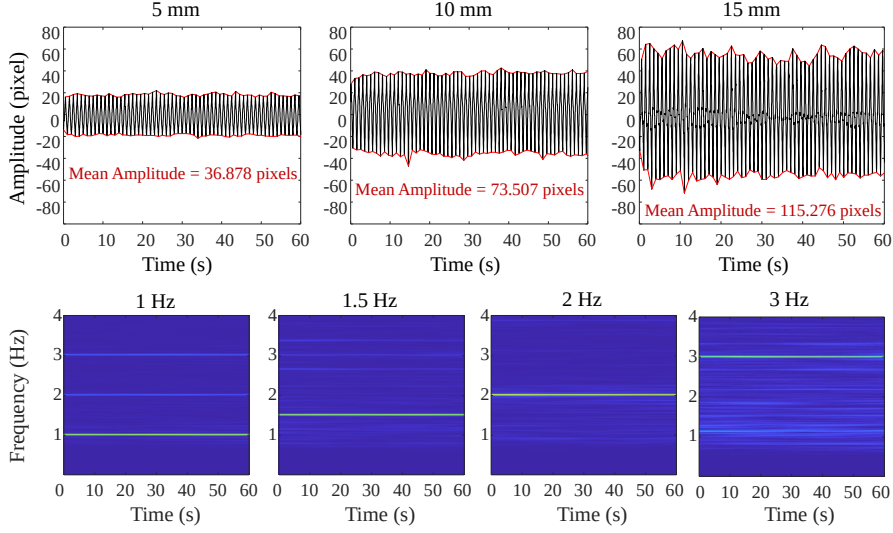


Figure 1: Preliminary testing experiment results. Top: The movement frequency is fixed, while the movement amplitude is adjusted to 5 mm, 10mm and 15mm; Bottom: The movement amplitude is fixed, while the movement frequency is adjusted to 1 Hz, 1.5 Hz, 2 Hz and 3 Hz.

4 Feasibility Testing Experiment

Prior to the formal acquisition of real cardiac motion data, a custom motion simulator is employed to preliminarily verify the validity of the spatial mapping relationship between the reconstructed motion by the proposed model and the actual motion depth. The motion simulator is driven by two square-wave pulse signals: one controls the motion direction (i.e., the motion frequency), while the other modulates the motion amplitude.

As illustrated in Fig. 1, the motion frequency is fixed and the amplitude is set to 5 mm, 10 mm, and 15 mm, respectively. The average peak-to-valley values of the reconstructed motion signals are approximately 36.878, 73.507, and 115.276 pixels, demonstrating a consistent proportional relationship with the actual motion amplitudes. Conversely, when the amplitude remains constant and the simulator is driven at different frequencies of 1 Hz, 1.5 Hz, 2 Hz, and 3 Hz, the dominant frequencies in the spectra of the reconstructed signals precisely match the actual motion frequencies. These results demonstrate that the proposed model is able to restore the amplitude relationships and frequency characteristics of the actual depth motion across varying motion amplitudes and frequencies, thereby providing preliminary validation of its feasibility for non-contact cardiac motion monitoring.

5 Waveform Morphology Analysis in a Controlled Laboratory Scenario

5.1 Analysis of changes before and after ice pack stimulation

In the data acquisition protocol, an ice pack is applied to the participant’s palm. Ice pack stimulation induces regulatory mechanisms in the autonomic nervous system, leading to changes in cardiac parameters and overall physiological state. As shown in Table 4, the reconstruction performance before and after ice-pack stimulation remains consistent across all types of signals (i.e., SCG, GCGy, and GCGx), with no significant differences observed (SCG: $r=0.642/0.641$, $S=92.818\%/93.179\%$;

GCGy: $r=0.503/0.507$, $S=88.927\%/88.910\%$; GCGx: $r=0.644/0.641$, $S=92.245\%/92.218\%$). Notably, the reconstructed SCG and GCGy signals before and after ice-pack stimulation outperform the raw speckle motion signals from both Camera 1 and Camera 2 in terms of correlation and similarity. These findings underscore the theoretical soundness of reconstructing cardiac 3D motion using a dual-camera system with distinct defocus levels, highlighting its intrinsic robustness even under moderate physiological perturbations.

Table 4: The Pearson correlation coefficients and similarity index between the reconstructed/raw speckle motion signal and reference signal in laboratory scenes.

Type / Signal	Pearson correlation coefficient (r)						Similarity index (S)					
	Position 1 (R-band Position)			Position 2 (G-band Position)			Position 1 (R-band Position)			Position 2 (G-band Position)		
	All ¹	Before ²	After ³	All	Before	After	All	Before	After	All	Before	After
Re-Signal ⁴	0.651	0.642	0.641	0.526	0.513	0.518	93.063%	92.818%	93.179%	91.994%	91.891%	91.955%
SCG Cam1-Rawx ⁵	0.339	0.324	0.338	0.322	0.314	0.314	83.792%	83.662%	84.014%	85.596%	85.317%	85.651%
Cam2-Rawx ⁶	0.344	0.329	0.342	0.325	0.316	0.317	84.236%	84.090%	84.486%	85.589%	85.328%	85.618%
Re-Signal	0.510	0.503	0.507	0.538	0.531	0.533	88.894%	88.927%	88.910%	89.232%	89.248%	89.243%
GCGy Cam1-Rawx	0.465	0.460	0.463	0.485	0.482	0.479	85.887%	86.082%	86.125%	87.773%	87.971%	87.935%
Cam2-Rawx	0.458	0.454	0.457	0.487	0.483	0.482	86.246%	86.498%	86.482%	87.709%	87.900%	87.898%
Re-Signal	0.648	0.644	0.641	0.497	0.494	0.488	92.270%	92.245%	92.218%	90.552%	90.543%	90.438%
GCGx Cam1-Rawy ⁷	0.628	0.624	0.623	0.477	0.475	0.468	91.743%	91.805%	91.693%	89.635%	89.666%	89.571%
Cam2-Rawy ⁸	0.650	0.647	0.644	0.473	0.469	0.464	92.083%	92.108%	92.043%	89.578%	89.625%	89.487%

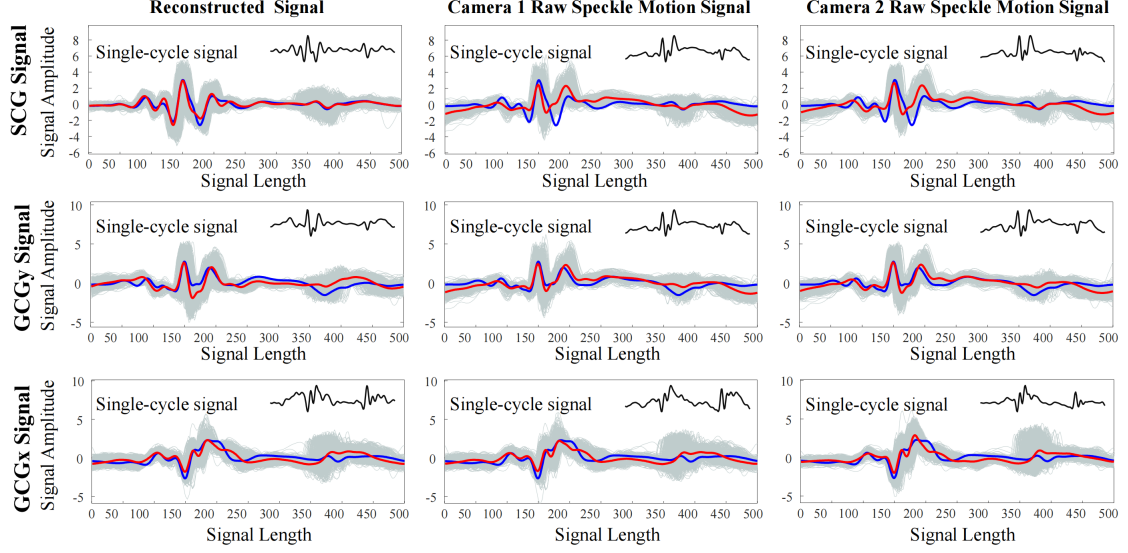
The boldface character denotes the best result. ¹All: All Data; ²Before: Data Before Ice Pack Stimulation; ³After: Data After Ice Pack Stimulation; ⁴Re-Signal: Reconstructed Signal; ⁵Cam1-Rawx: Camera 1 Raw Speckle X-direction Motion Signal; ⁶Cam2-Rawx: Camera 2 Raw Speckle X-direction Motion Signal; ⁷Cam1-Rawy: Camera 1 Raw Speckle Y-direction Motion Signal; ⁸Cam2-Rawy: Camera 2 Raw Speckle Y-direction Motion Signal.

5.2 Analysis of Reconstruction Performance at Different Laser Positioning Configurations

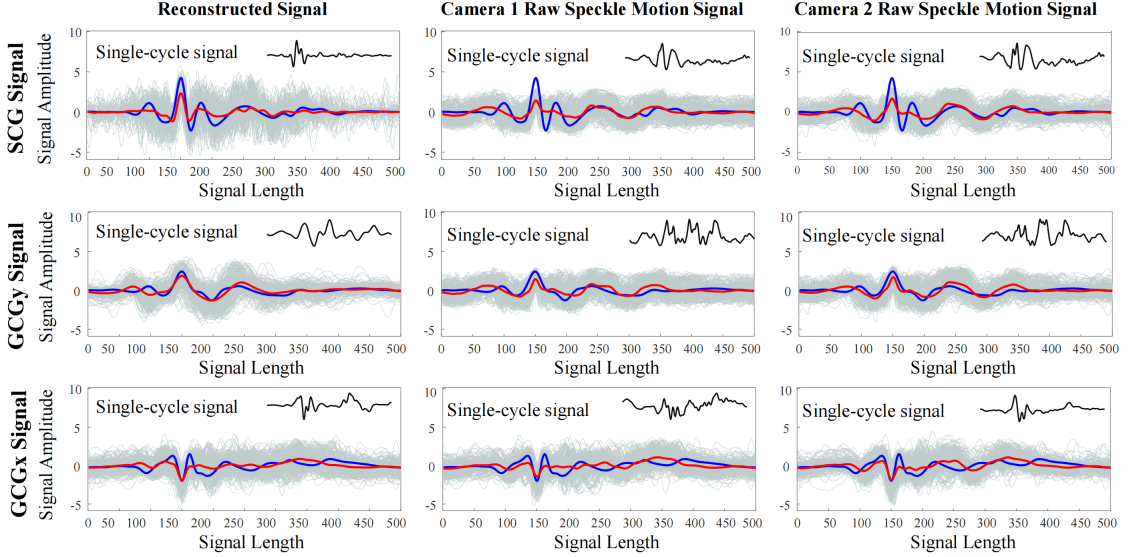
We obtained the raw speckle motion signal at two distinct positions (Position 1 and Position 2) to reproduce the results of the theoretical model. The results show a consistent performance trend across both measurement positions (see Table 4), i.e., the reconstructed SCG and GCGy signals consistently outperform the raw speckle motion signals from individual cameras, while the performance of the reconstructed GCGx signals remains largely comparable to that of the raw speckle motion signals. This comparable performance across different positions strengthens the theoretical foundation and underscores the feasibility of the proposed cardiac 3D motion reconstruction model under varying spatial configurations, further supporting its conceptual soundness and repeatability. Notably, slight performance differences are observed between reconstructed signals from Position 1 and Position 2, which may result from differences in the manifestation of cardiac 3D motion at different anatomical positions, as well as discrepancies in signal acquisition position (i.e., laser localization position) and reference positions (i.e., inertial sensor position).

6 Waveform Visualization in Laboratory and ICU Scenarios

(i) Laboratory Scenario: As shown in Fig. 2 (a), reconstruction performance is illustrated in both the controlled laboratory and ICU clinical scenarios. In SCG and GCGy of both scenes, the reconstructed signal (or raw speckle motion signal (X-direction)) has high similarity to the reference



(a) Illustrative example of result visualization under a controlled laboratory scenario.



(b) Illustrative example of result visualization under a real ICU clinical scenario.

— Reconstructed Signal/Camera Raw Speckle Motion Signal — Average Reference Signal
 — Average Reconstructed Signal/ Average Camera Raw Speckle Motion Signal

Figure 2: Examples of visualization of reconstruction performance under laboratory and ICU conditions, showing all signal cycles (gray), average reconstruction/raw speckle motion signal (red), average reference signal (blue), and representative single-cycle signal (black).

signal, but the reconstructed signal is closer to the reference signal in some waveform details. The performance of the reconstructed GCGx signal is comparable to that of the raw speckle motion signal (Y-direction), which is consistent with the theoretical modeling results described above.

(ii) ICU Scenario: Although the laboratory experiments validate the feasibility of the proposed cardiac 3D motion reconstruction framework, the controlled conditions pose relatively mild challenges to the model. Therefore, we collected data from ICU patients to further evaluate the framework’s capability in reconstructing cardiac 3D motion under pathological conditions characterized by more complex cardiac dynamics and physiological disturbances. Fig. 2 (b) presents an example of reconstruction results in the ICU setting. Compared to the laboratory results (Fig.

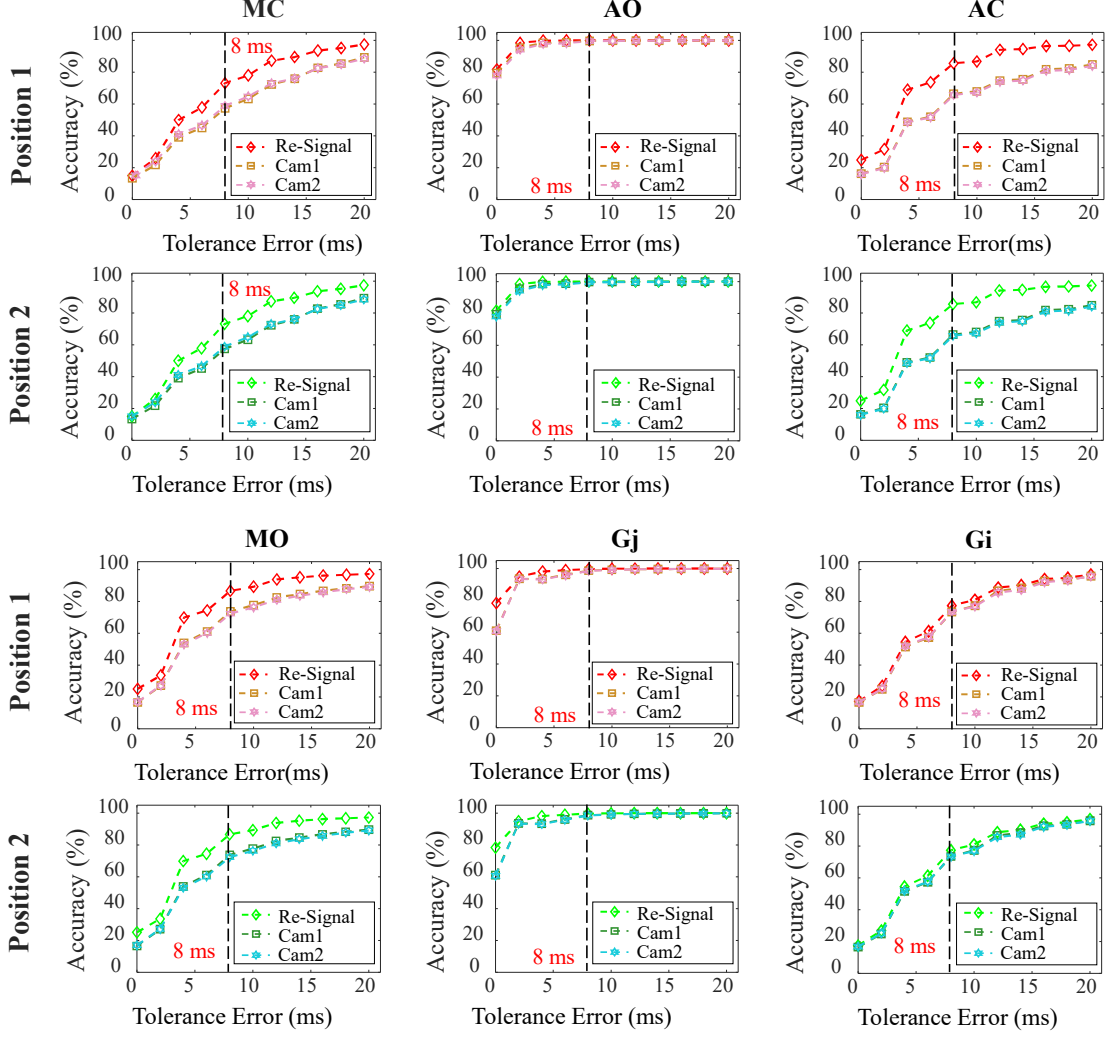


Figure 3: Localization accuracy of 6 biomarkers (AO, AC, MO, MC, Gj, and Gi) between the reconstructed/raw speckle motion signal and the reference signal with a tolerance error of 0 to 20 ms. The black dashed line indicates an allowable error of 8 ms (i.e., 2 sampling points of the signal before interpolation). Re-Signal: Reconstructed Signal; Cam1: Camera 1 Raw Speckle Motion Signal; Cam2: Camera 2 Raw Speckle Motion Signal.

2 (a)), the cycle-to-cycle consistency of all types of signals (indicated in gray) is substantially reduced in the ICU scenario, with markedly increased morphological variability between cardiac cycles. Furthermore, examples of individual cycle waveforms (in black) clearly demonstrate that key cardiac event markers (such as AC and MO) are often indistinct or absent in the cardiac motion signals of ICU patients. This is mainly attributed to the following factors: first, ICU patients exhibit pathological conditions (e.g., arrhythmias and weak cardiac function) and the injection of various medications (e.g., dexmedetomidine, bisoprolol, epinephrine and dobutamine) induces significant fluctuations in their intrinsic cardiac cycle signal morphology; second, involuntary convulsions and spasms due to brain injury, as well as the use of mechanical ventilation devices like respirators, affect the manifestation of cardiac pulsations on the thoracic surface transiently. The signal reconstruction is based on the raw speckle motion signals from the subject's chest, which, like the IMU, essentially monitors the movement of the subject's thoracic surface. The reference SCG and GCG signals are analyzed using the average signal of a recorded segment, which effectively

suppresses sporadic motion artifacts and random noise. Visually, the averaged signals in the ICU setting show that the reconstructed SCG and GCGy waveforms more closely resemble the reference signals in several details compared to the raw speckle motion signals, initially demonstrating the applicability and potential of the proposed model under pathological conditions.

7 Localization Accuracy of Six Biomarkers

Fig. 3 illustrates the localization accuracy of six key cardiac event markers at the two acquisition locations with a tolerance error of 0-20 ms. The results show that, under the same tolerance error, the reconstructed signals generally achieve higher localization accuracy than the raw speckle motion signals from a single camera. Specifically, AO and Gj are used for cycle segmentation, which results in high accuracy ($> 60\%$) even at a strict 0 ms tolerance error, and the performance is comparable between the reconstructed and single-camera signals. For other biomarkers in the SCG signal (i.e., MC, AC, and MO), the reconstructed signal demonstrates significantly higher localization accuracy. This improvement is primarily attributed to the fact that the single-camera speckle motion contains a coupled mixture of SCG and GCGy components, which can lead to shifts in the positions of these markers and reduce localization precision. In contrast, for the Gi marker derived from GCGx, the localization accuracy is comparable between the reconstructed and original signals, which is consistent with the results of the waveform morphology analysis.

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